YUZHEN CHEN

+1 734-548-0546 ◇ yuzhenchen@g.harvard.edu ◇ yuzhench.github.io ◇ @yuzhench

EDUCATION

Harvard University Cambridge, MA
M.S. in Computational Science and Engineering Expected May 2027
Relevant Coursework: Video Generation & Multi-modality World Model & VLA

University of Michigan Ann Arbor, MI
B.S.E. in Computer Engineering; **GPA:** 3.9/4.0 May 2025
Relevant Coursework: Data Structures & Algorithms, Machine Learning, Computer Vision, Computer Architecture, Robotic Kinematics and Dynamics, Robot Perception, SLAM, Linear Algebra, Multivariable Calculus, Differential Equations, Discrete Math, Probability and Statistics, Digital Circuit Design, Logic Design, Embedded System, GPU Parallel Computation, Robot 3D Reconstruction, Human Computer Interactive

HONORS & AWARDS

OMPP Summer Research Fellowship | Harvard University 2026
Best Paper Award, Runner-Up | CVPR 2026 WMAS Workshop 2026
Spotlight | CVPR 2026 3D-LLM/VLA Workshop 2026
Best Poster Award | IROS 2023 IPPC Workshop 2023
James B. Angell Scholar | University of Michigan 2023-2024
Dean's List | University of Michigan 2022-2025

RESEARCH EXPERIENCE

T-WLA: Bridging Slow Vision and Fast Touch [1]
Research Assistant, Advisors: Paul Pu Liang, Mengyu Wang, Antonio Torralba, and Wojciech Matusik 2026

- Developed a multi-rate world–language–action architecture that bridges low-frequency visual planning and high-frequency tactile correction through shared visual memory.
- Designed a lightweight tactile editor that uses cached task context and recent touch to asynchronously refine unexecuted actions without rerunning vision, supported by future-image and pressure prediction during training.
- Reduced finger-joint MPJPE by 30% on EgoTouch and 27% on Tachin relative to the best evaluated baselines; achieved 50% average success across four real-world dexterous manipulation tasks with 30 demonstrations per task, versus 23% for T-Rex.

GEM-4D: Geometry-Enhanced Video World Models for Robot Manipulation [2] Cambridge, MA
Research Assistant, Advisors: Dr. Paul Pu Liang (MIT Media Lab) & Dr. Mengyu Wang (Harvard AI and Robotics Lab) Sept. 2025 - Present

- Developed a geometry-grounded video world model that distills dense 4D correspondence supervision into a video diffusion transformer during training, improving physical consistency with no added inference cost.
- Proposed an adaptive inverse dynamics system that converts generated videos into executable 6-DoF robot trajectories using vision foundation models and confidence-gated pose recovery.
- Achieved state-of-the-art 4D scene generation—improving RGB video quality (SSIM 76.2→82.1, FVD 33.2→31.8) and point-cloud reconstruction (Chamfer distance 0.244→0.200) over the best prior baselines—and raised real-world robot manipulation success from 61% to 81%.

Spatially Calibrated Visuotactile Learning for Dexterous Manipulation
Ongoing Research 2026 - Present

- Built a sensor-calibrated capture system aligning multi-view video, hand positions, hand poses, and tactile measurements in a shared spatial frame, recording vision at 30 Hz and touch and pose at 120 Hz.
- Collected contact-rich demonstrations covering deformable objects, tool use, biochemical procedures, and occluded manipulation.
- Developing a tactile-conditioned world–action model that predicts latent world states to guide manipulation.

WCM: World-Cognition Model for Generalizable Human–Robot Interaction [3] Cambridge, MA
Research Assistant, Advisor: Kaichen Zhou Feb. - July 2026

- Designed WCM, a human-centered embodied agent built on the SLAK architecture (Sensing, Logic, Action, Knowledge) with an asynchronous runtime that lets reasoning, dialogue, and execution proceed concurrently, so the robot keeps acting instead of halting to think between steps.
- Introduced a human-in-the-loop teaching mode and a chain-of-thought distillation pipeline that refines both teaching episodes and autonomous rollouts into step-level CoT supervision, enabling the agent to acquire long-horizon tasks from user demonstration and improve continually.
- Achieved a 73.8% average success rate across a nine-task real-world human–robot interaction suite—including tasks held out from CoT fine-tuning and a long-horizon task learned purely through teaching—while cutting task time from 103.5 s to 38.5 s and GPU busy time from 103.5 s to 22.0 s against a VLA baseline.

Act to See: Evolving Articulated Memory for Closed-Loop Vision-Language Planning [4] Boston, MA
Research Assistant, Advisor: Dr. Mengyu Wang (Harvard AI and Robotics Lab) Jan. 2026 - Present

- Developed a VLM-guided manipulation framework that uses an evolving URDF as spatial memory, alongside task progress and interaction outcomes, to decide when to probe, manipulate, or stop.
- Built an interaction-driven loop that estimates joint parameters from observed part motion, fuses pose-aligned RGB-D geometry, and uses the updated model to generate subsequent linear or revolute robot trajectories.
- Achieved 100% valid joint reconstruction with 4.23 s processing time and 81.8% drawer / 72.7% door task completion in SAPIEN; validated on a UF850 robot with 5/5 drawer and 4/5 door successes in fixed setups.

Learning Ophthalmologist Clinical Reasoning for Glaucoma Diagnosis from Fundus Images [5] Boston, MA
Research Assistant, Advisor: Dr. Mengyu Wang (Harvard AI and Robotics Lab) Jan. 2026 - Present

- Developed a two-stage vision–language reasoning pipeline coupling a frozen RETFound encoder with MedGemma-27B: attention pooling and five task-specific heads extract structured visual evidence, which conditions six-step chain-of-thought (CoT) generation alongside the input image.
- Trained the evidence heads with uncertainty-weighted multi-task objectives and fine-tuned MedGemma-27B using LoRA and supervised CoT learning, with class-weighted next-token loss computed only on response tokens and normalized per sample.
- Built a five-fold out-of-fold evidence-generation and report-adaptation pipeline to reduce training leakage and expose the reasoning model to prediction errors; achieved 94.7% balanced accuracy, 94.8% precision, and 0.874 BERTScore-F1 against expert reports.

Multi-Modal Semantic Perception in Robotic Navigation [9], [10] Ann Arbor, MI
Research Assistant, Advisor: Dr. Ram Vasudevan Apr. 2023 - May 2025

- Modified Kinect Fusion to integrate material-based color mapping into 3D reconstruction, enabling real-time visualization of surface friction coefficients.
- Optimized material classification model using MiT-B1 architecture, achieving 4x faster inference (9 FPS) while maintaining accuracy across 6 material classes.
- Developed a voting pipeline combining SAM and SegFormer models, significantly reducing high-noise issues at object junctions in real-time material predictions.

Learning Physical Object Properties from Images Using Diffusion Model [7] Ann Arbor, MI
Research Assistant, Advisor: Dr. Arpan Kusari Sept. 2024 - May 2025

- Developed an automated pipeline using Meta MatSynth and Replica datasets to generate large-scale training data with varied material properties, lighting conditions, and camera angles.
- Trained an encoder to extract material properties (base color, metallic, roughness, reflectance, anisotropy, transmission) from images, enabling accurate Physically-Based Rendering function generation.
- Implemented Monte-Carlo differentiable rendering as decoder, achieving robust image reconstruction.

Conformalized Reachable Sets for Obstacle Avoidance With Spheres (CROWS) Ann Arbor, MI
Research Assistant, Advisor: Dr. Ram Vasudevan Apr. 2024 - Sept. 2024

- Developed ML models for CROWS, a safe motion planning system using conformalized neural networks for sphere parameter prediction, reducing planning time from 0.51s to 0.21s compared to ARMTD.
- Generated extensive training datasets through simulation-based sampling of motion trajectories, enabling robust training of three distinct neural networks for center, radii, and gradient predictions.
- Validated system effectiveness through successful collision-free motion demonstrations in both simulation and hardware experiments, ensuring safe robot operation in unstructured environments.

Real-Time Modeling for Elastic Rods [6], [8] Ann Arbor, MI
Research Assistant, Advisor: Dr. Ram Vasudevan Nov. 2023 - Sept. 2024

- Used DEFORM, a physics-based modeling system for real-time simulation of elastic rods, achieving accurate prediction of rope configurations in 3D space based on physical parameters.
- Implemented a motion capture pipeline for collecting ground truth data, facilitating model training, validation, and performance optimization through real-world rope behavior analysis.
- Applied the simulation framework to industrial robotics, enabling automated wire manipulation tasks including insertion, removal, and assembly operations.

SLIM-VDB: Real-Time 3D Probabilistic Semantic Mapping [11] Ann Arbor, MI
Research Assistant, Advisor: Dr. Katie Skinner May 2024 - Sept. 2024

- Contributed to a real-time 3D probabilistic semantic mapping system that leverages OpenVDB for memory-efficient volumetric scene representation.
- Integrated closed-set and open-set semantic predictions with RGB-D / point-cloud observations through probabilistic fusion to generate interpretable 3D semantic maps.
- Built visualization and evaluation tools for benchmark mapping experiments, supporting analysis of semantic quality, integration time, and memory efficiency.

INTERNSHIPS

Bardor | Software & System Engineer May - Sept. 2025

- Bridging software/hardware requirements between Detroit clients and Chinese manufacturers, securing projects for instrument clusters and infotainment displays across Chrysler, Jeep, and Polaris platforms.

UMich Shapiro Design Lab | Educational Technology Developer May 2024 - May 2025

- Designed and built a multi-screen dice with six independent displays that respond to motion, enabling dynamic group learning in language education and serving as a platform for game design exploration - deployed at university language centers and gaming design courses, presented at Serious Play Conference 2025.

PROJECTS

Embodied Intelligence Robot – ROVER | Independent Project May. 2025 - Present

- Engineered a high-performance 5-DOF robotic arm achieving 1-meter reach with 1kg payload capacity, delivering 3x better cost-effectiveness (<$2000) than commercial alternatives
- Achieved 50ms control loop latency through optimized ROS2 architecture and custom motion planning algorithms, enabling real-time multi-modal interaction with RGB-D SLAM, LiDAR, and natural language processing
- Served as a testing platform for research teams, accelerating VLA/VLM model deployment cycles through providing affordable, reproducible physical validation beyond simulation

PUBLICATIONS

* denotes equal contribution.

1. Zhou, K.*, **Chen, Y.***, Yao, H.*, Jiang, C., Fu, R., Yu, X., Dai, W., Kabbarah, B., Chen, H., Goluguri, I., Dou, Z., Chen, H., Zhan, F., Wang, M., Matusik, W., Torralba, A., Liang, P.P. (2026). T-WLA: Bridging Slow Vision and Fast Touch. *arXiv preprint.* **(*Equal contribution)**
2. Zhou, K.*, **Chen, Y.***, Zhan, F., Hua, H., Chen, G., Chang, X., Qu, A., Du, Y., Liu, Z., Liang, P.P., Wang, M. (2026). GEM-4D: Geometry-Enhanced Video World Models for Robot Manipulation. *ECCV 2026.* **(*Equal contribution; Best Paper Runner-Up & Spotlight, CVPR 2026 Workshops)**
3. **Chen, Y.**, Zhou, K. (2026). WCM: World-Cognition Model for Generalizable Human–Robot Interaction. *RSS 2026 Workshop on Human-centric Mobile Manipulation.*
4. Liu, Y.*, **Chen, Y.***, Wang, B.*, Li, C., Lu, J., Li, H., Li, Y., Ji, M., Zhou, K., Wang, M., Zhan, F. (2026). Act to See: Evolving Articulated Memory for Closed-Loop Vision-Language Planning. *Manuscript.* **(*Equal contribution)**
5. Zhou, K.*, **Chen, Y.***, Yildiz, E.*, Shi, M., Dai, D., Chen, G., Zheng, J., Wang, H., Zhan, F., Saini, C., Shen, L.Q., Guo, Y., Liang, P.P., Wang, M. (2026). Learning Ophthalmologist Clinical Reasoning for Glaucoma Diagnosis from Fundus Images. *medRxiv preprint* (under review at a journal).
6. Chen, Y., Sang, F., Peng, S., Wu, X., Zong, Y., **Chen, Y.**, Li, A., Zhang, B., Vasudevan, R. (2026). DEFT: Differentiable Branched Discrete Elastic Rods for Modeling Furcated DLOs in Real-Time. *The International Journal of Robotics Research (IJRR)*, accepted.
7. **Chen, Y.**, Son, H., Kusari, A. (2025). MatPredict: A Dataset and Benchmark for Learning Material Properties of Diverse Indoor Objects. *arXiv preprint* (under review).
8. Chen, Y., Zhang, Y., Brei, Z., Zhang, T., **Chen, Y.**, Wu, J., Vasudevan, R. (2024). Differentiable Discrete Elastic Rods for Real-Time Modeling of Deformable Linear Objects. *CoRL 2024*, Munich.
9. Ewen, P., Gunjal, G., Chen, H., Li, A., **Chen, Y.**, Vasudevan, R. (2023). Multi-Modal Semantic Perception Using Bayesian Inference. *IROS 2023 IPPC Workshop*, Detroit. **(Best Poster Award)**
10. Ewen, P., Chen, H., **Chen, Y.**, Li, A., Bagali, A., Gunjal, G., Vasudevan, R. (2024). You've Got to Feel It To Believe It: Multi-Modal Bayesian Inference for Semantic and Property Prediction. *RSS 2024.*
11. Sheppard, A., Ewen, P., Wilson, J., Sethuraman, A.V., Adewole, B., Li, A., **Chen, Y.**, Vasudevan, R., Skinner, K.A. (2026). SLIM-VDB: A Real-Time 3D Probabilistic Semantic Mapping Framework. *IEEE RA-L.*
12. Oral Presentation: **Chen, Y.**, Cai, B. (2023) Development of high-performance but low-cost sensing node for intelligent monitoring system. *U-M UROP Symposium*, Ann Arbor, MI. **(Best Poster Award)**

LEADERSHIP POSITIONS

Junior Organizer | PhysWorldAI: 1st Workshop on Physical World AI, NeurIPS 2026 — 2026
Robotics Community Event Organizer | MIT CHIEF — 2026 - Present
Teaching Assistant | EECS442 Computer Vision; Physics 140X — Sept. 2022 - Apr. 2025
Departmental Ambassador | ENGR 110; UMich Robotics Department — Sept. 2023 - Dec. 2024
Student Tutor | UMich Engineering Center for Academic Success — Jan. - Apr. 2023

SKILLS

Programming Languages: Python, C++, C, CUDA, MATLAB, JavaScript, Shell Scripting, LaTeX
ML/DL Frameworks: PyTorch, TensorFlow, JAX, TensorRT, ONNX, Hugging Face, OpenCV, scikit-learn, NCNN
Robotics & Vision: ROS/ROS2, Gazebo, MoveIt, PCL, Open3D, SLAM (ORB-SLAM, LIO-SAM), Motion Planning (OMPL, ARMTD)
ML Models & Algorithms: Transformers (BERT, GPT, Whisper), Vision Models (YOLO, SAM, DINO), Diffusion Models, Neural Rendering (NeRF, 3D Gaussian Splatting), Reinforcement Learning (PPO, SAC)
Tools & Platforms: Docker, Git, Conda, WandB, TensorBoard, CUDA Toolkit, AWS/GCP, Weights & Biases
3D/CAD Software: Fusion360, Blender, SolidWorks, MeshLab, CloudCompare, Mimics
Languages: Mandarin Chinese (Native), English (Fluent)